

# Erdős Problem #657

EP-657

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

## ORIGINAL PROBLEM STATEMENT

Is it true that if  $A \subset \mathbb{R}^2$  is a set of  $n$  points such that every subset of 3 points determines 3 distinct distances (i.e.  $A$  has no isosceles triangles) then  $A$  must determine at least  $f(n)n$  distinct distances, for some  $f(n) \rightarrow \infty$ ?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>  
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>